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EXPT.NO - 7b

TITLE - 0/1 Knapsack problem using Greedy Method

In [2]:

```python
# Taking inputs from user
n = int(input("Enter the number of items: "))

print("Enter the weight and profit for each item:")
weight = []
profit = []

for i in range(n):
    w = int(input("Weight of item {}: ".format(i + 1)))
    p = int(input("Profit of item {}: ".format(i + 1)))
    weight.append(w)
    profit.append(p)

capacity = int(input("Enter the capacity of the Knapsack: "))

def greedy_knapsack(capacity, weight, profit, n):
    # Compute profit-to-weight ratio for each item
    ratio = [(profit[i] / weight[i], weight[i], profit[i]) for i in range(n)]

    # Sort items by ratio in descending order
    ratio.sort(reverse=True, key=lambda x: x[0])

    total_profit = 0
    current_weight = 0

    for ratio_value, w, p in ratio:
        if current_weight + w <= capacity:
            total_profit = total_profit + p
            current_weight = current_weight + w
        else:
            remaining_capacity = capacity - current_weight
            total_profit = total_profit + ratio_value * remaining_capacity
            break

    return total_profit

max_profit = greedy_knapsack(capacity, weight, profit, n)

print("==============================================")
print("Maximum Profit Earned (Greedy Approach) =", max_profit)
print("==============================================")
```

```
Enter the number of items: 3
Enter the weight and profit for each item:
Weight of item 1: 10
Profit of item 1: 60
Weight of item 2: 20
Profit of item 2: 100
Weight of item 3: 30
Profit of item 3: 120
Enter the capacity of the Knapsack: 40
==============================================
Maximum Profit Earned (Greedy Approach) = 200.0
==============================================
```

```
In [ ]:
```